

Ischemic Stroke

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Aim of the Session

The aim of this session is to highlight how the basic principles of neuroanatomy, pathology and physiology relate to common neurological conditions, particularly stroke, and how these principles aid our understanding, diagnosis and treatment of these conditions.

Learning Outcomes :

Discuss the important pathophysiological principles of common neurological conditions

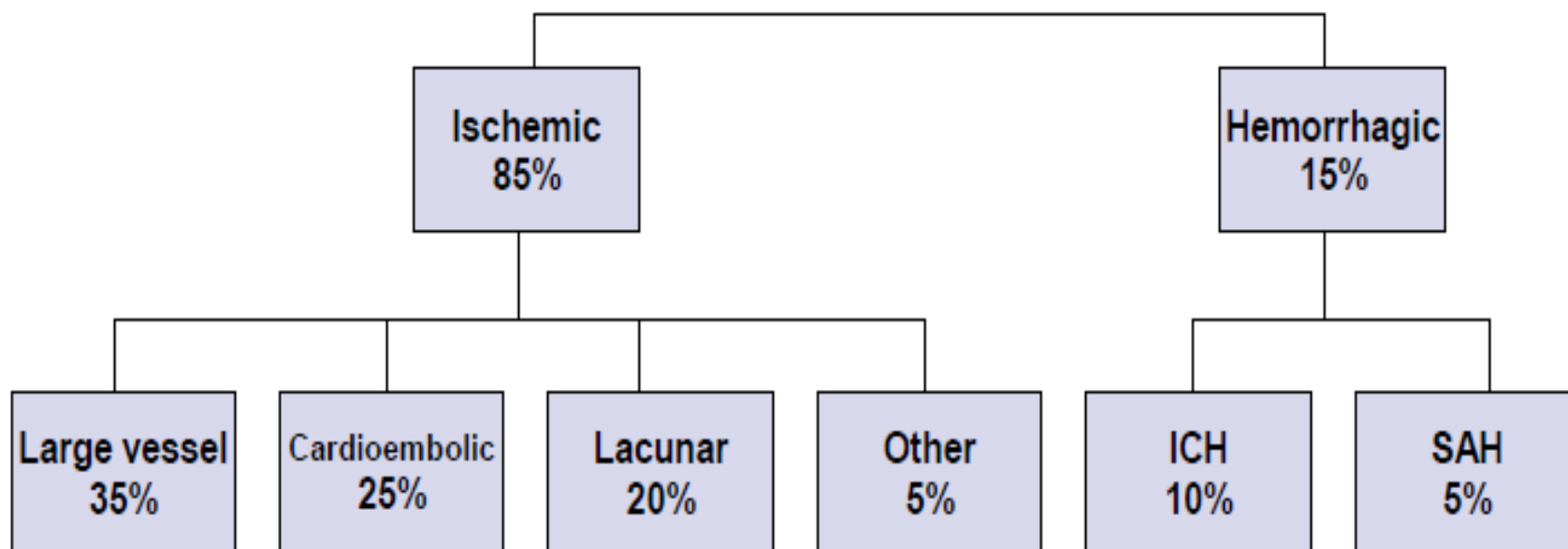
- Identify appropriate lines of enquiry in the history and physical examination of a patient with a neurological condition
- Define the clinical and pathological entity of stroke and describe its classification, pathophysiology and risk factors
- Describe how the clinical presentation following a stroke is determined by the neuroanatomical structures affected
- Discuss the importance of the early evaluation and management in cases of stroke
- Discuss, in broad outline, current stroke treatment strategies

Definition

- Stroke is a clinical syndrome of sudden focal or global cerebral dysfunction lasting more than 24 hours, of presumed vascular origin.



Classification of stroke in general



Anatomy of cerebral circulation

- ❖ 20% of cardiac output—

- ❖ 750-1000ml/min

- ❖ >30 second interruption— neurologic metabolism is altered; metabolism stops in 2 minutes; brain cell death < 5 mins.



❖ **Anterior: Carotid Arteries** – middle & anterior cerebral arteries

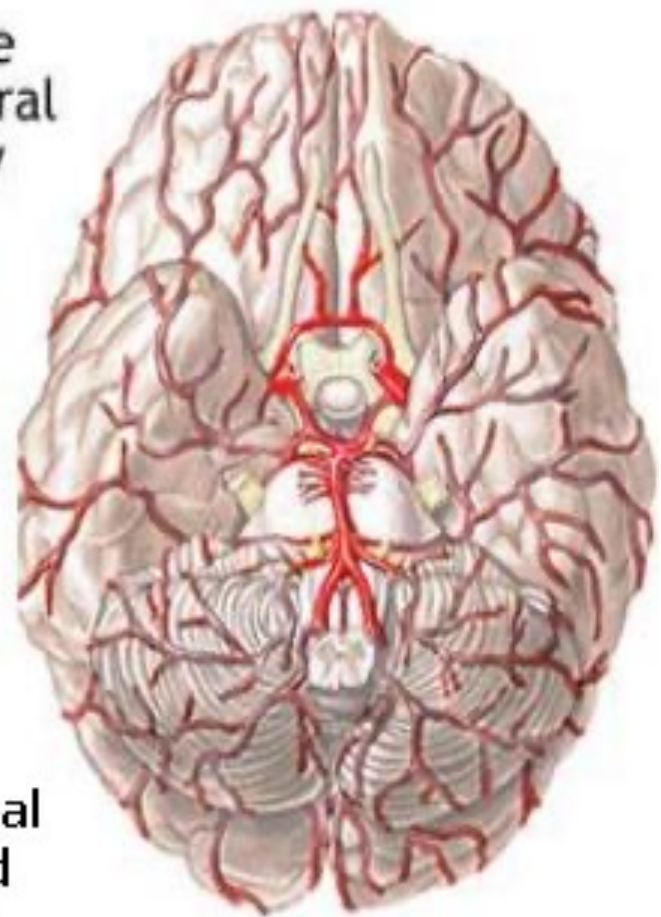
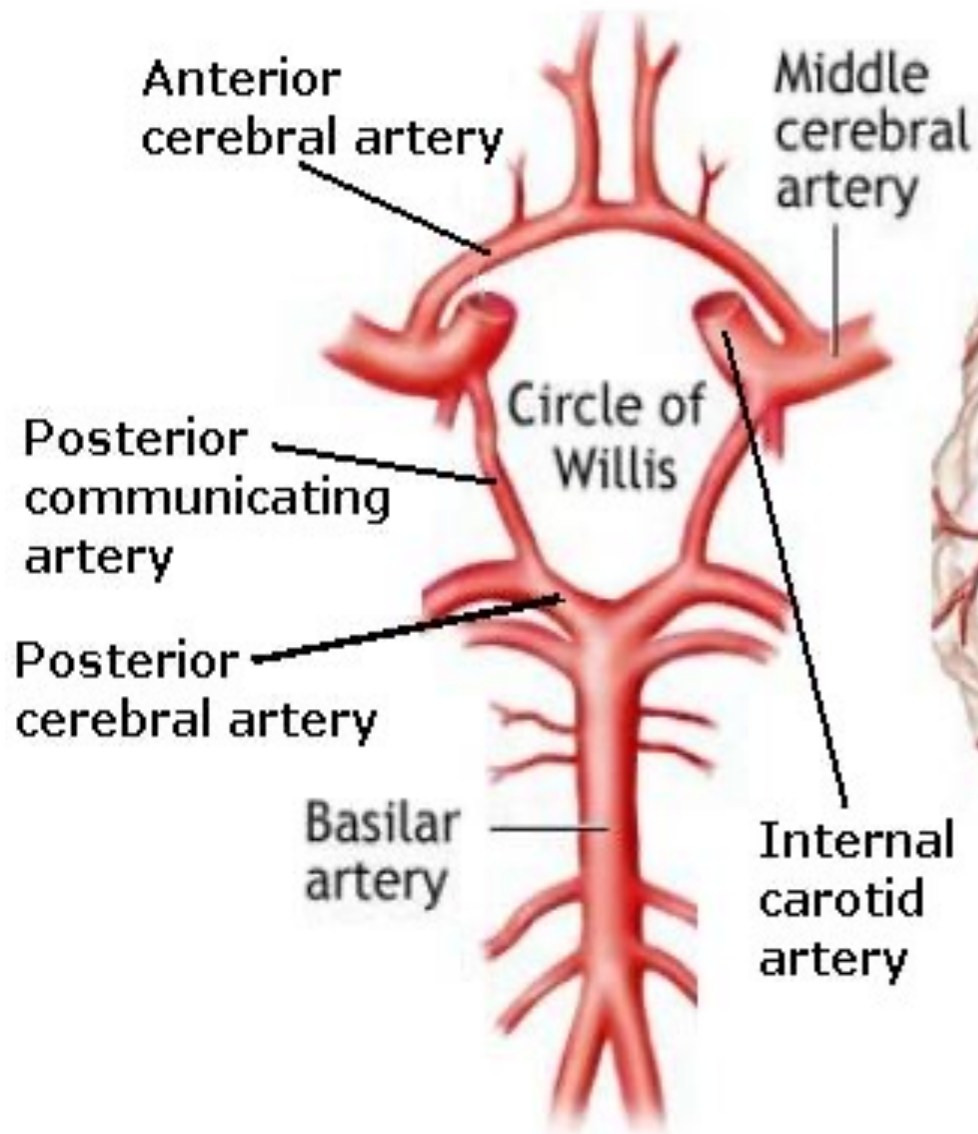
- ❖ frontal, parietal, temporal lobes; basal ganglion; part of the diencephalon (thalamus & hypothalamus)

❖ **Posterior: Vertebral Arteries** – basilar artery

- ❖ Mid and lower temporary & occipital lobes, cerebellum, brainstem, & part of the diencephalon

❖ **Circle of Willis** – connects the anterior & posterior cerebral circulation





Bottom view of brain



Pathophysiology of ischemic stroke

❖ **Atherosclerosis:** major cause of ischemic stroke

❖ **Thrombus formation & emboli development**

- ❖ Abnormal filtration of lipids in the intimal layer of the arterial wall
- ❖ Plaque develops & locations of increased turbulence of blood - bifurcations
- ❖ Increased turbulence of blood or a tortuous area
- ❖ Calcified plaques rupture or fissure
- ❖ Platelets & fibrin adhere to the plaque
- ❖ Narrowing or blockage of an artery by thrombus or emboli



Pathophysiology of ischemic stroke

- Thrombo-embolic occlusion of blood flow triggers a sequence of events, the ischemic cascade.
- Failure of energy production leads to anaerobic glycolysis and lactic acidosis, and failure of the ion pumps results in neuronal depolarisation and intracellular calcium overload.
- These events lead to the release of neurotoxic substances such as excitatory neurotransmitters (chiefly glutamate), inflammatory mediators (eg. prostaglandins, leukotrienes), toxic free radicals (eg. nitric oxide) and activated lytic enzymes (lipases, proteases), ultimately causing neuronal death.



- Ischemic damage depends on the degree and the duration of ischemia. Following complete occlusion of a vessel, a central core of densely ischemic tissue is irreversibly damaged (infarction) within minutes.
- Around the infarction is an area of critical ischemia (ischemic penumbra), which is inadequately perfused.



- Neurons in the penumbra are energy deficient and electrically quiescent, but have intact ion pumps and are viable.
- Prolonged ischemia will lead to their death and extension of the infarct.
- If perfusion is restored within a certain period of time, neurons in the penumbra can be salvaged; this constitutes a therapeutic window of opportunity.
- In humans, this window period is believed to be 3 to 6 hours.(4.5)



Common Warning Signs of Stroke

- | Sudden weakness or numbness of the face, arm, and leg on one side of the body
- | Loss of speech, or trouble speaking or understanding speech
- | Dimness or loss of vision, particularly in only one eye or half of both eyes
- | Sudden onset of blurred or double vision
- | Unexplained dizziness
- | Sudden onset of unsteadiness, lack of coordination, difficulty walking, or falling
- | Sudden excruciating headache
- | Recent change in personality or mental abilities, including memory loss



FAST

F face

A arm

S speech

T is time which is hyperacute onset

Stroke Recognition:

3 Steps to Stroke Recognition



Ask the person to smile
and stick out tongue



Ask the person to make
a complete sentence



I can't fall
tell side
which one.



© MILLER



Ask the person to
raise both arms.



Contact someone if the person cannot
perform these 3 steps!



Risk factors

❖ Nonmodifiable:

- ❖ **Age** – Occurrence doubles each decade >55 years
- ❖ **Gender** – Equal for men & women; women die more frequently than men
- ❖ **Race** – African Americans, Hispanics, Native Americans, Asian Americans -- higher incidence
- ❖ **Heredity** – family history, prior transient ischemic attack, or prior stroke increases risk



Controllable Risks with Medical Treatment & Lifestyle Changes:

High blood pressure

Cigarette smoking

High blood cholesterol

Heart Disease

Oral contraceptive use

Sickle cell disease

Asymptomatic carotid stenosis

Hypercoagulability

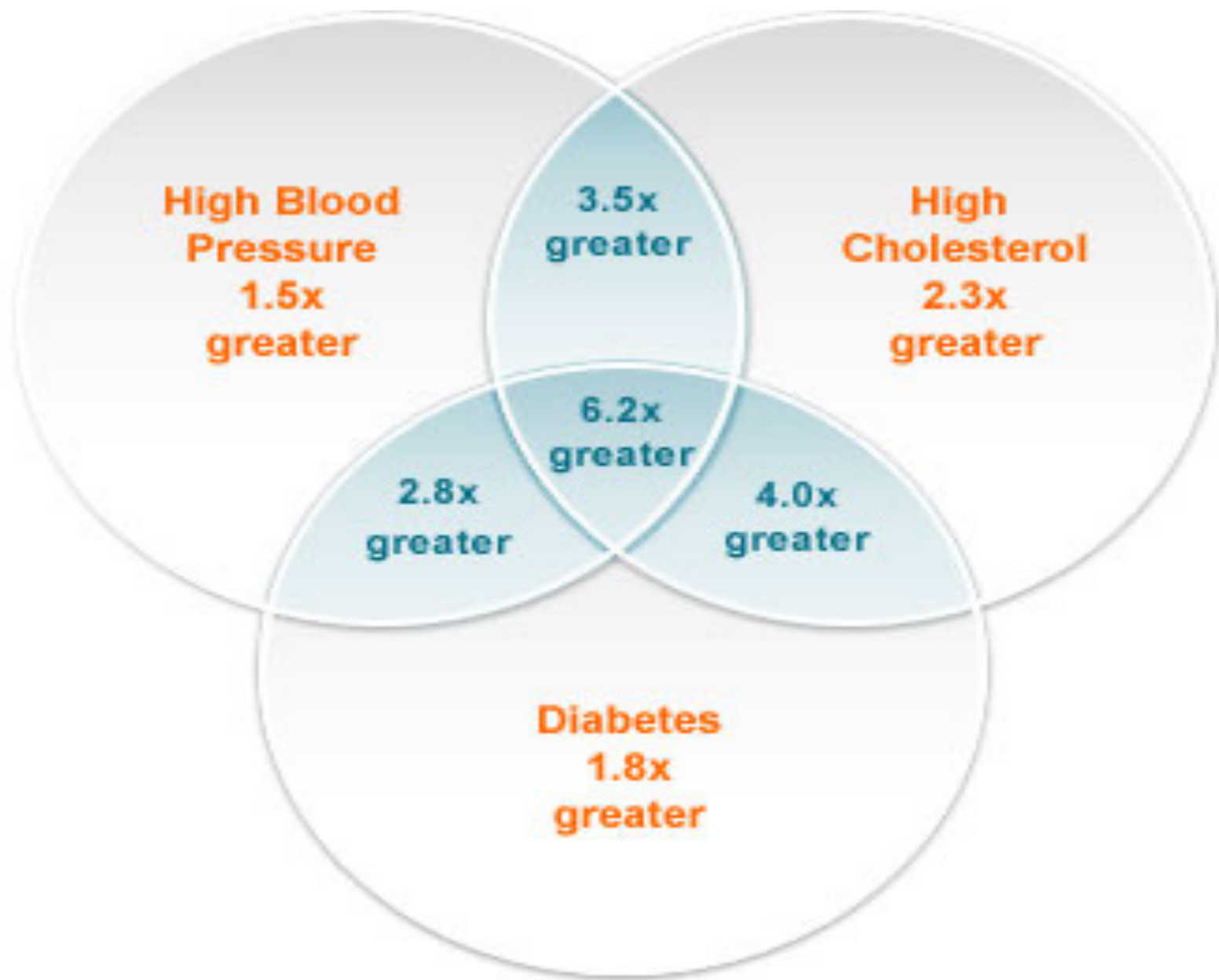
Diabetes

Obesity

Atrial fibrillation

Physical inactivity





Transient Ischemic Attack

- ❖ Temporary focal loss of neurologic function
- ❖ Caused by ischemia of one of the vascular territories of the brain
 - ❖ Microemboli with temporary blockage of blood flow
- ❖ Lasts less than 24 hrs – often less than 15 mins
- ❖ Most resolve within 3 hours
- ❖ Warning sign of progressive cerebrovascular disease



Types of ischemic stroke

❖ **Ischemic Stroke**—inadequate blood flow to the brain from partial or complete occlusions of an artery--85% of all strokes

- **Extent of a stroke depends on:**

- Rapidity of onset
- Size of the lesion
- Presence of collateral circulation

- Symptoms may progress in the first 72 hours as infarction & cerebral edema increase



1. Thrombotic Stroke

- ❖ Lumen of the blood vessels narrow – then becomes occluded – infarction
- ❖ Associated with HTN and Diabetes Mellitus
 - ❖ >60% of strokes
 - ❖ 50% are preceded by TIA

2. Embolic Stroke

- ❖ Embolus lodges in and occludes a cerebral artery
- ❖ Second most common cause of stroke – 24%
- ❖ Emboli originate either in the heart – atrial fibrillation, MI, infective endocarditis, rheumatic heart disease, valvular prostheses **Or** from large arteries (carotid)
- ❖ Any age group
- ❖ Recurrence common if underlying cause not treated



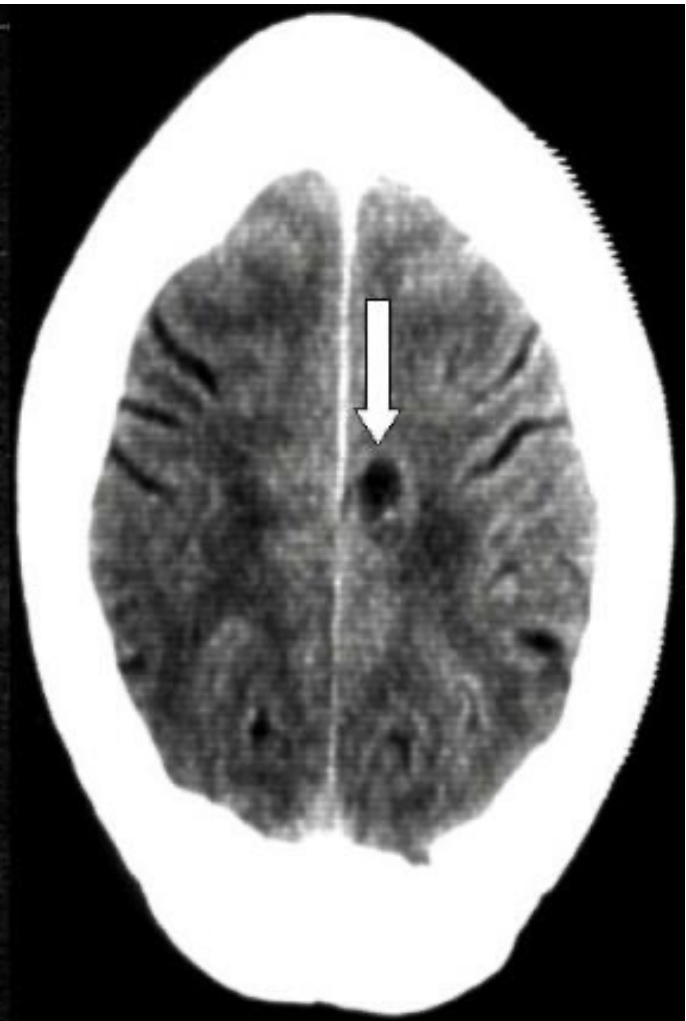
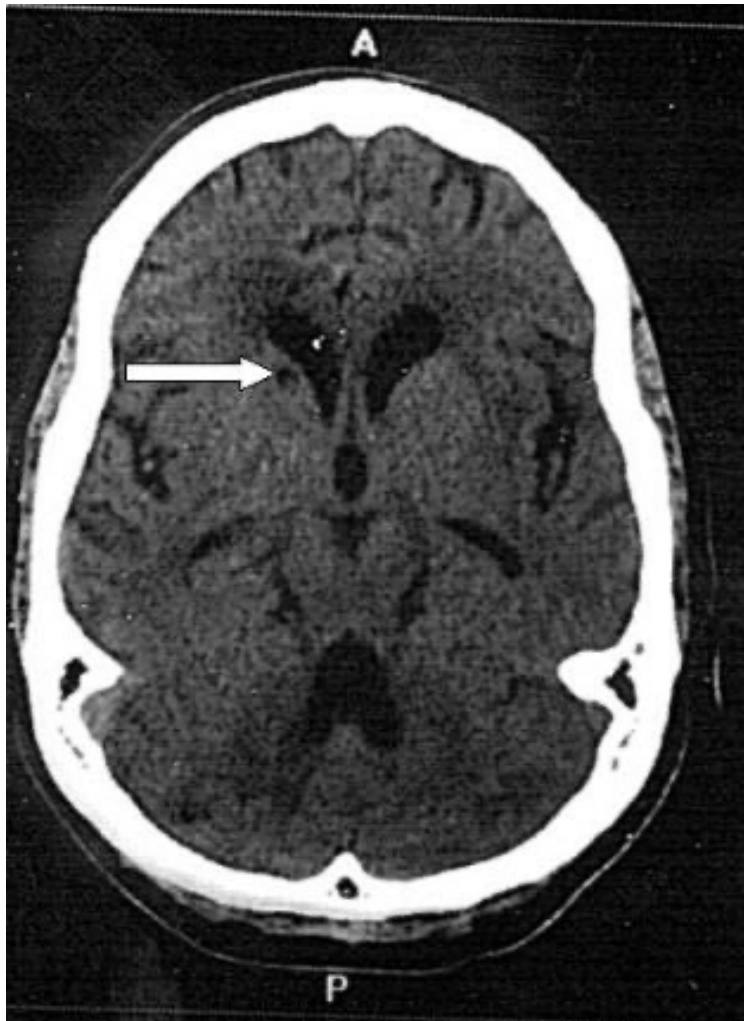
Lacunar infarction

- A restricted area of infarction (2-20 mm) due to ischemia in the territory of a single perforating artery.
- Lacunar strokes constitute a quarter of all ischemic strokes
- generally attributed to lipohyalinosis.



- **pure motor stroke/hemiparesis**
(most common lacunar syndrome: 33-50%)
- **Ataxic hemiparesis**
(second most frequent lacunar syndrome)
- **Dysarthria/clumsy hand**
(sometimes considered a variant of ataxic hemiparesis, but usually still is classified as a separate lacunar syndrome)
- **Pure sensory stroke**
- **Mixed sensorimotor stroke**





Management

1. Recognize the stroke patient

- History
- Clinical examination
- Exclude stroke mimic

(70 years old man, hypertensive , diabetic presented with sudden onset right sided weakness and mouth deviation).

2. Is it ischemic or hemorrhagic

- Non-contrast CT of the Head is initial imaging study of choice and readily available. It is very sensitive for blood in the acute phase but not sensitive for acute ischemic stroke nearly 100% sensitive by 7 days.



- May shows early signs of ischemia in the 1st 3 hours
 - loss of gray/white matter distinction
 - hypodensity
 - mass effect, edema
 - hyperdense middle cerebral artery sign
- Re-evaluate the time of symptom onset, if early signs of ischemia are present





3. Detect the time of onset

- 3- 4.5 hours is a good candidate for thrombolytic therapy
- More than 4.5 hours not treated by it

4. Acute treatment rather than thrombolytic therapy

- Anti platelet
- ant lipid (statin)
- Blood pressure monitoring



5. Prevention and treatment of acute complication

- Raised intracranial pressure
- Aspiration pneumonia
- DVT
- Bed sore
- Cardiac arrhythmia

6. Rehabilitation

- early mobilization and physiotherapy is a must

7. Secondary prevention

- Work up and investigations
- Life style modification
- Risk factor management



Thrombolytic therapy

- All patients with disabling acute ischemic stroke who can be treated within 4.5 hours after symptom onset should be evaluated *without delay* to determine their eligibility for treatment with t-PA.
- All eligible patients should receive intravenous alteplase (t-PA) within one hour of hospital arrival
 - door-to-needle time < 60 minutes Alteplase rTPA
- 0.9mg /Kg, 10% of total dose –Bolus 2-3 mins and 90% of total dose –Infuse over 60 mins



The Art of t-PA Decision Making

Treat Enthusiastically

Early

Young

Glucose, BP normal

On Protocol

Moderate-Severe Strokes

Good CT – higher ASPECTS

Treat nervously and selectively (if at all)

Late

Old

↑↑Glucose, ↑↑BP

Off Protocol

Minor Stroke

Bad CT – ASPECTS < 3

Dual antiplatelet therapy

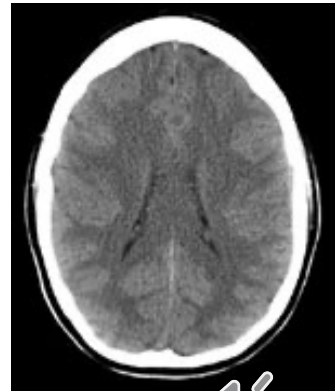


- Thrombus is formed during ischemic stroke.
- Alteplase binds to fibrin in a thrombus:
 - converts plasminogen to plasmin
 - initiates local fibrinolysis with minimal systemic effects.
- Alteplase is cleared rapidly from circulating plasma by the liver.
 - >50% cleared within 5 min after infusion
 - 80% cleared within 10 min



Inclusion Criteria

- Acute ischemic stroke with disabling neurological deficits
- Acute ischemic stroke presenting within 4.5 hours of stroke symptom onset.
- No hemorrhage on CT scan



Exclusion Criteria:

Absolute Contraindications:

- Intracranial hemorrhage
- Active internal bleeding
- Endocarditis or acute pericarditis

*If any of the above criteria are present, **t-PA should NOT be administered**. T-PA is “absolutely contraindicated” in the above circumstances.



Physician Name _____

Physician Signature _____

Acute Stroke Inclusion Criteria

- | Yes | No | |
|--------------------------|--------------------------|--|
| ☐ | ☐ | Acute ischemic stroke with disabling neurological deficit |
| ☐ | ☐ | Treatment can be initiated within 4.5 hours of symptom onset |
| ☐ | ☐ | Patient Evaluated by Stroke Neurologist, ER Physician and approved for rtPA |

Absolute Contraindications (DO NOT TREAT)

- | Yes | No | |
|--------------------------|--------------------------|------------------------------------|
| ☐ | ☐ | Intracranial hemorrhage |
| ☐ | ☐ | Active internal bleeding |
| ☐ | ☐ | Endocarditis or acute pericarditis |

Relative Contraindications (STOP AND RETHINK)

- | Yes | No | |
|--------------------------|--------------------------|--|
| ☐ | ☐ | Platelets less than 100,000 mm ³ |
| ☐ | ☐ | INR greater than 1.7 or PTT greater than 40 |
| ☐ | ☐ | Glucose less than 2.7 or greater than 22.2 mmol/L |
| ☐ | ☐ | BP greater than 185/110 after 2 attempts to reduce BP |
| ☐ | ☐ | Surgery or trauma within the last 14 days |
| ☐ | ☐ | Stroke or head injury within preceding 3 months |
| ☐ | ☐ | Pre-treatment CT scan showing: <ul style="list-style-type: none"> • Hypodensity, which could represent evolving infarction greater than 4.5 hours old • Tumor, aneurysm, AV malformation |
| ☐ | ☐ | Seizure at stroke onset |
| ☐ | ☐ | History of respiratory, retroperitoneal, GI or Urinary hemorrhage within past 21 days |
| ☐ | ☐ | Lumbar puncture or arterial puncture at a non-compressible site within the last 7 days |
| ☐ | ☐ | Intracranial surgery or intraspinal surgery less than 2 months |
| ☐ | ☐ | Previous history of CNS bleeding |
| ☐ | ☐ | Serious underlying medical illness, or serious surgical risk (i.e. abdominal aortic aneurysm) |
| ☐ | ☐ | Patient is pregnant (requires negative pregnancy test) |
| ☐ | ☐ | Recent Myocardial Infarction |

Date and Time _____

Physician Name
(Please Print) _____

Physician Signature _____



Prior to Infusion of t-PA:

- Blood pressure management

Maintain SBP < 185mmHg and DBP < 110mmHg

BP Treatment:

- Labetalol 10-20mg IV push over 1-2 min, repeat q10-20 min prn (max 300mg).
Do NOT use β -blockers if HR < 60bpm
- Hydralazine 10-20mg IV push over 1 min q20 min prn
- Blood sugar above 400 and below 40 should be treated before treatment
- Antiplatelet should be given after 24 hours of thrombolytic

*Among patients who are candidates for treatment with intravenous t-PA, attention to management of blood pressure is critical before, during, and after t-PA administration. (Grossman and Messereli; Drug Saf;1998;1999-122). If blood pressure can be lowered safely with limited use of antihypertensive agents, patient may be eligible for treatment. The stability of BP before starting t-PA should be assessed.

An elevated BP that requires continuous infusion (ie Nitroprusside) may not be sufficiently stable for patient to receive t-PA (APSS, 2008) (Adams et. al: Stroke: 2007; 38: 1655-1711). Please refer to the APSS education module, 'Hypertension Management in Acute Stroke' for additional information related to hypertension.

Excessively high BP is associated with an increased risk of hemorrhagic transformation.



- The main complication of thrombolytic therapy is bleeding, which may be intracranial or extracranial.
- Spontaneous hemorrhagic transformation, a recognised complication of cerebral infarction, may be aggravated by thrombolytic therapy.
- Angioedema is another common complication.



Acute treatment rather than thrombolytic therapy

Antiplatelet

- Aspirin 300 mg loading dose and 100 mg daily thereafter
- After 24 hours of thrombolytic
- Any patient arriving hospital out of window more than 4.5 hours



Statin therapy

- Atorvastatin 80 mg or reusovastatin 40 mg daily
- Close monitoring of liver enzyme

Blood pressure monitoring

Should not be treated unless

- Systolic >220mmHg or
Diastolic >120mmHg with 2 consecutive readings
- Organ failure (renal, heart failure and retinopathy)

Abrupt fall in BP may affect cerebral perfusion pressure



- Blood glucose content is another critical determinant of outcome after stroke.
- Both hypoglycaemia and hyperglycaemia can be detrimental.
- Hypoglycaemia should be corrected with dextrose infusions.
- Hyperglycaemia is associated with aggravation of cerebral edema and increased mortality, and needs treatment with insulin.



Treatment of complications

- Early detection and treatment of complications are essential to improve outcome after a stroke.
- **Cerebral edema**, or brain swelling, is probably the most important early complication.
- Treatment of cerebral edema should ideally be guided by intracranial pressure monitoring.
- In the absence of such facilities, patients with alteration of consciousness, large infarcts, and evidence of mass effect (midline shift, compression of ventricles) on CT scanning should be treated.



Management of raised intracranial pressure after stroke

- Elevate head end by 30°
- Avoid or correct aggravating factors _ hypoxia, hyperglycaemia
- Moderate fluid restriction
- Avoid hypoosmolar fluids eg. dextrose
- Giving osmotic agents (eg iv. mannitol) as indicated
- Hyperventilation
- IV barbiturates
- NB. Steroids are of no value



Other complications of stroke include

- hemorrhagic transformation,
- seizures
- respiratory and urinary infection
- deep vein thrombosis and pulmonary embolism
- acute peptic ulcer
- pressure sores
- neuropsychiatric sequelae such as anxiety or depression
- musculo-skeletal sequelae such as contractures, spasticity and adhesive capsulitis of the shoulder.



To prevent these complications we have to do

- Bedside swallowing test

To detect the dysphagia, if it is positive we should avoid oral feeding and NG tube should be put to prevent aspiration pneumonia.

- Low molecular weight heparin

preventive dose should be given within 48 hours to hemiplegic patient with pneumatic stocks to prevent DVT and pulmonary embolism.



- Gastric preventive measure (PPI or cimetidine) and pneumatic bed to prevent bed sore.
- Early detection of depression and treatment of it improve outcome of stroke.
- Patients with stroke are usually old and may have associated co-morbid conditions. These may interfere with the rehabilitation process (eg. chronic lung disease), or the treatment (eg. aspirin or warfarin in peptic ulcer disease), and need assessment and treatment in their own right.



Rehabilitation

- Rehabilitation is the process by which patients after a stroke are restored to their previous functional, mental and social capacity. This is best carried out by a multidisciplinary stroke team with the active participation of patients and care givers.



Secondary prevention

- Work up and investigations
- Life style modification
- Risk factor management



For secondary prevention of stroke you should answer

- What is the risk factor?
- Is the stroke embolic or thrombotic?

CBC, Lipid profile, RBS and HbA1c, RFT, LFT should be done to find the risk factors.

- To detect the source of embolization we should do full cardiac and large artery assessment.



- ECG to detect arrhythmia and ischemic heart disease
- Echocardiography to detect valvular heart disease, wall hypokinetic, intraventricular thrombus and cardiomyopathy
- Sometime Holter monitoring of heart to detect paroxysmal AF
- Carotid artery Doppler ultrasound to detect the stenosis and mobile thrombus.



- After full assessment we should plan for treating the patient

1. Life style modification

- Stop smoking
- Exercise
- Decrease weight
- Decrease salt intake

2. Blood pressure control using ARB and ACEI

Target is 130/80 to 150/90 according to the age and major organ function



- Blood sugar control
- Target of HbA1c is below 7 but even below 8 is acceptable in elderly to prevent hypoglycemia
- Anti lipid (statin) should be continued

Target LDL level should be below 100 mg/dl generally and below 70 in patient with multiple risk factors.

- Anti platelet (aspirin and clopidogril) if stroke is thrombotic and anti coagulant (warfarin and novel oral anti coagulant) if the stroke is embolic.
- If carotid artery stenosis is more than 50%, we should plan for carotid intervention by carotid artery stenting or endarterectomy.

